

The classical cube-maximal subcase of two rescaled-space conjectures

Run `fa.banach.001`, lane 1

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Status

Literature-implied answer (proper subcase). Lerner’s 2025 theorem proves both conjectures for Banach function spaces on $\mathbb{R}^n$ and the usual Hardy–Littlewood maximal operator over cubes. The source states the conjectures for a more general quasi-Banach/abstract-basis setting, which is not covered. Thus this is neither a new proof nor a full answer to the source formulations.

Source questions

In Z. Nieraeth, *Extrapolation in general quasi-Banach function spaces*, arXiv:2211.03458, Conjectures 2.38 and 2.39 occur together on PDF page 26. For an r -convex, s -concave function space X , write

$$X_{r,s} = \left[[(X^r)']^{(s/r)'} \right]'$$

Conjecture 2.38 says, for a basis $\mathcal{E}$ having the A_1 self-improvement property and under boundedness of $M^{\mathcal{E}}$ on $[(X^r)']^{(s/r)'}$, that

$$M^{\mathcal{E}} : X_{r,s} \rightarrow X_{r,s} \iff M^{\mathcal{E}} : X^r \rightarrow X^r.$$

Here the source permits X to be quasi-Banach. Conjecture 2.39, for Banach X , says that simultaneous boundedness on $X_{r,s}$ and $(X_{r,s})'$ is equivalent to simultaneous boundedness on X^r and $(X')^{s'}$.

Later theorem and exact match

A. K. Lerner, *A note on the maximal operator on Banach function spaces*, J. Geom. Anal. 35 (2025), Article 97, arXiv:2404.15671, explicitly restates these as Conjectures 1.1 and 1.2. Lerner assumes

$$1 \leq r < s \leq \infty, \quad X \text{ a Banach function space over } \mathbb{R}^n,$$

and takes M to be the standard cube Hardy–Littlewood maximal operator. Section 4, PDF pages 10–11, proves both conjectures. In particular:

- under boundedness of M on $[(X^r)']^{(s/r)'}$, boundedness on $X_{r,s}$ is equivalent to boundedness on X^r ;
- boundedness of M on both $X_{r,s}$ and $(X_{r,s})'$ is equivalent to boundedness on both X^r and $(X')^{s'}$.

These statements are exactly the source conjectures after specializing $\Omega = \mathbb{R}^n$, $\mathcal{E}$ to cubes, $r \geq 1$, and X to a Banach function space. The identification is explicit in Lerner’s introduction, which also notes that Nieraeth’s formulations are more general.

Why this does not settle the abstract version

The restriction is substantive. Lerner’s proof uses the local median maximal operator and a cube-based criterion for dual maximal boundedness. It also uses maximal self-improvement, the Fefferman–Stein inequality, A_p -regularity, and reverse-Hölder properties of Euclidean cube weights. The second conjecture further passes through a theorem turning two A_1 -regular convexifications into A_q -regularity. Nieraeth’s sole hypothesis that an abstract basis has the A_1 self-improvement property does not supply these dual and A_p -regularity mechanisms. The quasi-Banach part of Conjecture 2.38 also lies outside Lerner’s Lorentz–Luxembourg and Köthe-duality argument.

Accordingly, the correct status is a complete affirmative answer in the canonical Euclidean Banach setting and an unresolved remainder for general bases and the quasi-Banach range.

References

- [1] Z. Nieraeth, *Extrapolation in general quasi-Banach function spaces*, J. Fourier Anal. Appl. 29 (2023), Article 70; arXiv:2211.03458.
- [2] A. K. Lerner, *A note on the maximal operator on Banach function spaces*, J. Geom. Anal. 35 (2025), Article 97; arXiv:2404.15671.